

## Stage 20: Production Hardening, Observability, Security Audit & Final Deployment Readiness

### 1. Executive Summary

Stage 20 represents the final milestone of the **20-Stage Production Upgrade** for the **Alight International Multi-Vendor Marketplace**. This stage validates platform stability, database schema integrity across 46 Flyway migrations, end-to-end audit logging, security compliance, Prometheus/Actuator metrics, container readiness, and comprehensive cross-module test verification.

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| ALIGHT INTERNATIONAL MARKETPLACE PLATFORM |                         |                    |                   |
|-------------------------------------------|-------------------------|--------------------|-------------------|
| [Auth & RBAC]                             | [Multi-Tenant Vendors]  | [Catalog & Facets] | [Cart & Pricing]  |
|                                           |                         |                    |                   |
| - Session Revocation                      | - Bank & Address Config | - Full-Text Search | - Saved For Later |

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| [Order & Sub-Orders]  | [Multi-Gateway Payments] | [Logistics & Shipping] | [Returns & RMA]     |
|-----------------------|--------------------------|------------------------|---------------------|
| - Sub-Order Splitting | - Multi-Currency FX      | - AWB Gen & Milestones | - Reverse Logistics |
|                       |                          |                        |                     |

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| [Coupons & Flash]    | [Reviews & Q&A]          | [Notifications & Help] | [Settlements & Tax] |
|----------------------|--------------------------|------------------------|---------------------|
| - Proportional Split | - Verified Purchase Gate | - Omnichannel In-App   | - 1% GST TCS / TDS  |
|                      |                          |                        |                     |

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|-----------------------------------|
| [AUDIT LOGS & PROMETHEUS METRICS] |
|-----------------------------------|

## 2. Platform Architecture & Production Infrastructure

### 2.1 Technology Stack & Versions

- : Java 21 LTS, Spring Boot 3.3.4, Spring Security 6 (JWT / stateless auth), Spring Data JPA / Hibernate 6, Liquibase / Flyway.
- : PostgreSQL 16 with JSONB indexing and UUID primary keys (`gen_random_uuid()`).
- : Redis for session cache and distributed locks; asynchronous Spring Events.

- `actuator` : Spring Boot Actuator ( `/actuator/health` , `/actuator/metrics` , `/actuator/prometheus` ), Micrometer, SLF4J MDC structured logging.
  - `build` : Maven 3.9+, Docker multi-stage builds ( `backend/Dockerfile` , `docker-compose.prod.yml` ).
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### 3. Complete 20-Stage Verification Matrix

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| Stage | Domain Module          | Primary Capabilities Verified                             | Test Suite Class                            |
|-------|------------------------|-----------------------------------------------------------|---------------------------------------------|
|       | Architecture Baseline  | Monolith structure, Maven build, SLF4J MDC, domain events | Baseline Config                             |
| 2     |                        |                                                           | DatabaseIntegrityIntegrationTest            |
|       | RBAC & Permissions     | 5 Roles (ADMIN, VENDOR, CUSTOMER, SUPPORT, LOGISTICS)     | AuthServiceTest                             |
| 4     |                        |                                                           | HealthControllerTest                        |
|       | Modular Monolith       | Decoupled domain module packaging and event bus           | Module Structure                            |
| 6     |                        |                                                           | AuthAndSessionIntegrationTest               |
|       | Vendor Multi-Tenancy   | Vendor onboarding, KYC verification, store approval       | VendorManagementIntegrationTest             |
| 8     |                        |                                                           | ProductCatalogIntegrationTest               |
|       | Search & Facets        | Dynamic facet aggregation, price ranges, brand filters    | SearchAndFacetIntegrationTest               |
| 10    |                        |                                                           | InventoryAndStockReservationIntegrationTest |
|       | Cart & Saved-For-Later | Multi-vendor cart totals, saved-for-later move            | CartAndSavedForLaterIntegrationTest         |
| 12    |                        | ORD-V1                                                    | OrderManagementIntegrationTest              |
|       | Payments & Escrow      | Multi-gateway adapters, webhook deduplication, FX         | PaymentAndMultiCurrencyIntegrationTest      |
| 14    |                        |                                                           | LogisticsAndShippingIntegrationTest         |
|       | Returns & RMAs         | Customer RMA initiation, QC inspection, restocking        | ReturnsAndRmaIntegrationTest                |
| 16    |                        |                                                           | CouponsAndPromotionsIntegrationTest         |
|       | Reviews & Q&A          | Verified purchase reviews, rating cache sync, Q&A         | ReviewsRatingsAndQaIntegrationTest          |
| 18    |                        |                                                           | NotificationsAndSupportIntegrationTest      |
|       | Settlements & Ledger   | Escrow release, 1% GST TCS, payout UTI, tax ledgers       | SettlementAndPayoutsIntegrationTest         |

| Stage | Domain Module | Primary Capabilities Verified | Test Suite Class                                   |
|-------|---------------|-------------------------------|----------------------------------------------------|
| 20    |               |                               | ProductionHardeningAndObservabilityIntegrationTest |

4. Verification Results

- : 64 test suites
- : BUILD SUCCESS (0 compiler warnings / errors)
- : 46 Flyway scripts ( V1 through V46 )